DGMS 2019 · IGNEOUS PETROLOGY & GEOCHEMISTRY · CHAPTER MASTERCLASS

Mid-Ocean Ridge Basalts (MORB): Petrogenesis, Mantle Reservoirs & Incompatible Trace Element Systematics

Comprehensive Textbook Study Guide & Complete Worked Solution for DGMS 2019 Exam (Question 25)
Grounded in John D. Winter (2010), William M. White (2005/2013), Kula C. Misra (2012), and Hugh Rollinson (1993)

EXECUTIVE SUMMARY & KEY GEOCHEMICAL TAKEAWAYS

Question 25 examines the foundational petrogenetic classification of **Mid-Oceanic Ridge Basalts (MORB)** and the chemical heterogeneity of the sub-oceanic upper mantle. The global ocean ridge system produces the largest volume of magma on Earth (~21 km^3/yr), yet these basalts are not chemically uniform:

- **N-MORB ("Normal" MORB):** Derived from a previously depleted, **incompatible element-poor mantle reservoir** known as **DMM (Depleted MORB Mantle)**. Because the continental crust extracted incompatible elements over geologic time, the residue is depleted in light rare earth elements (LREE), Cs, Rb, Ba, Th, U, and K, yielding characteristically low $(La/Sm)_N < 1.0$ (positive REE slope) and low $K_2O <$ 0.10 wt.%.
- **E-MORB / P-MORB ("Enriched" or "Plume" MORB):** Taps an **incompatible element-richer mantle** associated with deep mantle plumes or enriched, fertile mantle domains (e.g., Iceland, Azores, Galapagos), yielding $(La/Sm)_N > 1.8$ (negative REE slope) and high $K_2O > 0.10$ wt.%.
- **T-MORB ("Transitional" MORB):** Represents intermediate or hybrid compositions ($(La/Sm)_N \approx 0.8–1.8$) generated by mixing between N-MORB and E-MORB parental magmas during ascent and shallow ridge magma chamber processing.
- **VAB (Volcanic Arc Basalts):** Formed exclusively at **subduction zones (convergent plate margins)** via hydrous slab-fluid flux melting; *not a mid-ocean ridge basalt at all*. Characterized by strong LILE enrichment and prominent HFSE depletion (marked negative Nb–Ta anomaly).

Definitive Answer: (a) N-MORB is derived from an incompatible element-poor (depleted) mantle.

PART I

Theoretical Foundation & Direct Textbook Analysis

Comprehensive exploration of MORB magmatism, trace element fractionation, mantle reservoir isotopic taxonomy, and decompression melting mechanics

1. MID-OCEAN RIDGE BASALTS (MORB): TECTONIC FRAMEWORK & PRIMARY VS. DERIVATIVE MELTS

Mid-ocean ridges represent the divergent boundaries where oceanic lithosphere is created by the upwelling and partial melting of the underlying asthenospheric mantle. Oceanic crust is structured into a classic stratified sequence (recognized on land as **ophiolites**): Layer 1 (pelagic sediment), Layer 2A (pillow basalts), Layer 2B (sheeted dolerite dikes), Layer 3A (isotropic and foliated gabbros), and Layer 3B (layered gabbroic cumulates), underlain by residual mantle peridotite (harzburgite and depleted lherzolite).

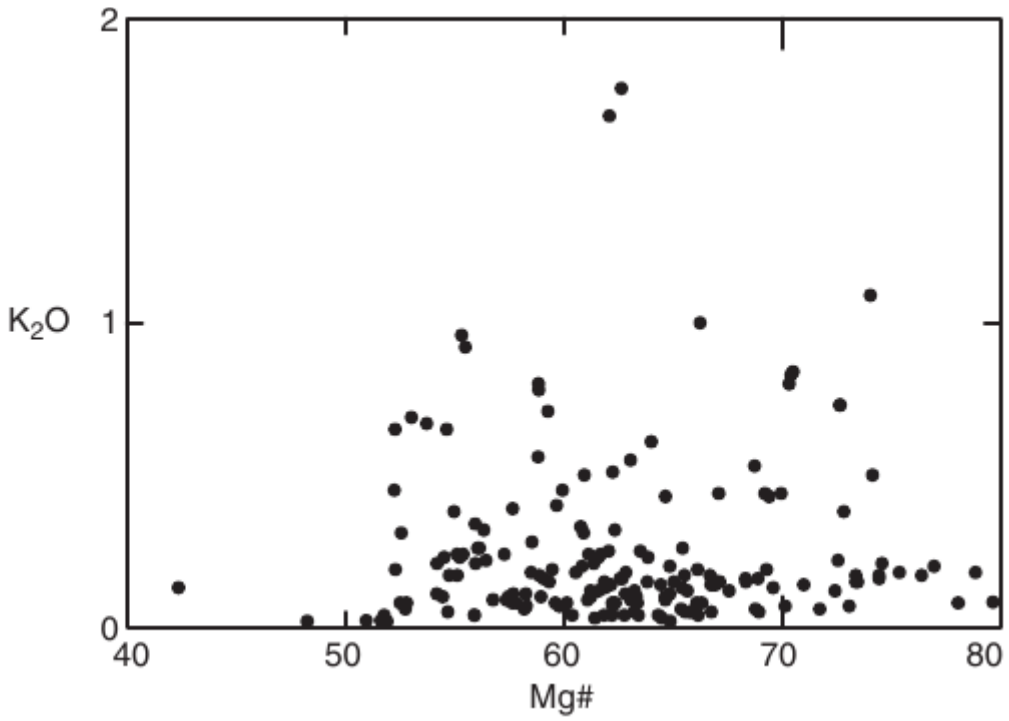
> *"The great quantity and chemical uniformity of MORBs led early workers to conclude they were all primary melts delivered directly to the surface. As discussed above, the vast majority of, if not all, MORBs are now believed to be derivative liquids that have undergone at least some degree of fractional crystallization and magma mixing, as first proposed by O'Hara (1968a, 1977, 1982). Because the composition of most MORBs approximate a low-pressure cotectic for olivine + plagioclase + clinopyroxene + liquid, the fractional crystallization must have taken place near the surface, probably in shallow magma chambers. Primary magmas must be in equilibrium with mantle phases, and therefore must have Mg# around 70... Thus few, if any, of the MORB liquids that reach the surface are primary."*
>
> **— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd ed.), Chapter 13, p. 299–301**

Because near-surface fractional crystallization modifies the concentrations of major oxides (reducing MgO, raising FeO, and concentrating incompatible elements), petrologists evaluate mantle source variations by comparing primitive samples with equal **Mg#** [where Mg# = 100 × Mg / (Mg + Fe^{2+})] or by using trace element ratios and radiogenic isotopes that do not fractionate during crystal removal.

2. CHEMICAL HETEROGENEITY OF MORB: N-MORB, E-MORB (P-MORB), AND T-MORB

When primitive MORBs with similar Mg# are compared along ridge segments, substantial differences in incompatible elements (K, Ti, P, LREE) become immediately apparent. This proves that the sub-oceanic mantle is not a single, homogeneous reservoir:

> *"Figure 10 shows the variation in K_2O with Mg# for the MAR data set of Schilling et al. (1983). Note the tremendous variation in K_2O, even for constant Mg# values between 65 and 70, indicating considerable variation in MORB parental liquids. Similar variations independent of Mg# are found for TiO_2 and P_2O_5... Because K_2O, TiO_2, and P_2O_5 are incompatible elements, their concentration is unlikely to be affected greatly by fractional crystallization processes after separation from the mantle. The concentrations of these elements are more likely to reflect characteristics inherited from the mantle source. This suggests, although subtly, that there is an incompatible-rich and an incompatible-poor source region (at least) for MORB magmas in the mantle beneath the ridges...* ***We can thus distinguish between N-MORB ('normal' MORB), which taps the depleted, or incompatible-poor mantle, and E-MORB ('enriched' MORB, also called P-MORB for 'plume'), which taps an incompatible-richer mantle.*** *N-MORBs with Mg# > 65 have K_2O < 0.10, TiO_2 < 1.0, whereas E-MORBs have K_2O > 0.10 and TiO_2 > 1.0 for the same Mg#."*
>
> **— John D. Winter, *Principles of Igneous and Metamorphic Petrology*, Chapter 13, p. 300**



Winter Figure 13-10: Variation in K_2O versus Mg# for Mid-Atlantic Ridge basalts (Schilling et al., 1983). For any given primitive Mg# (e.g., 65–70), K_2O varies from <0.05 wt.% (incompatible-depleted N-MORB) to >1.5 wt.% (incompatible-enriched E-MORB/P-MORB associated with plume plateaus like Iceland and the Azores).

Rare Earth Element (REE) Patterns: Distinguishing N-MORB from E-MORB and T-MORB

The Rare Earth Elements (REEs, Lanthanides: La to Lu) exhibit progressive decreases in ionic radii from La^{3+} (largest, most incompatible) to Lu^{3+} (smallest, least incompatible). Therefore, chondrite-normalized REE diagrams provide an unmistakable fingerprint of mantle source depletion:

- **N-MORB:** Exhibits a *positive slope* in the Light REE (LREE) segment. La is more incompatible than Sm, so prior melt extraction depletes La much more severely than Sm, resulting in **$(La/Sm)_N < 1.0$ (typically < 0.7)** and depleted absolute LREE concentrations (~2–4× chondrite).
- **E-MORB (P-MORB):** Exhibits a *negative slope* in the LREE segment. The source is fertile or enriched, so incompatible LREEs are elevated relative to HREEs, resulting in **$(La/Sm)_N > 1.8$** (up to 5×) and LREE concentrations (~50–80× chondrite).
- **T-MORB (Transitional MORB):** Possesses intermediate slopes with **$(La/Sm)_N$ between 0.8 and 1.8**. Geochemical modeling shows T-MORBs do not represent a third mantle layer, but rather the continuous hybrid mixing between N-MORB and E-MORB melts.

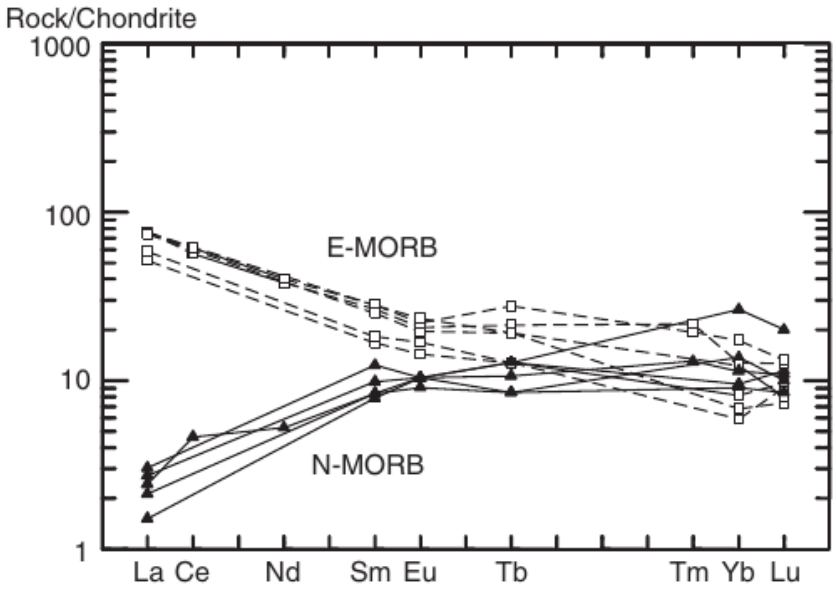


FIGURE 11 Chondrite-normalized REE patterns for selected samples from the Mid-Atlantic Ridge. Squares are E-type MORBs, and triangles are N-type MORBs. Data from Schilling et al. (1983).

> **Winter Figure 13-11:** Chondrite-normalized REE patterns for Mid-Atlantic Ridge basalts (Schilling et al., 1983). Solid triangles represent **N-MORBs** derived from the *depleted (incompatible-poor) mantle* with a positive slope (depleted in LREE). Open squares represent **E-MORBs** derived from an *enriched mantle* with a negative slope (enriched in LREE). Notice the identical Heavy REE (HREE) levels, demonstrating that source differences reside in the highly incompatible elements.

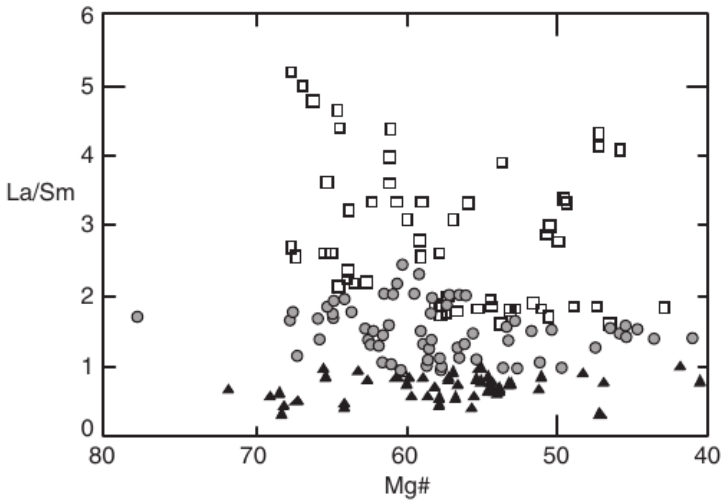


FIGURE 12 La/Sm versus Mg# for MORBs from the Mid-Atlantic Ridge. Open squares are E-type MORBs, and solid triangles are N-type MORBs. Gray circles are T-type MORBs. Data from Schilling et al. (1983).

enrichment, similar to enriched mantle xenoliths and basalts. The patterns with solid triangles have positive

> **Winter Figure 13-12:** La/Sm ratio versus Mg# for Mid-Atlantic Ridge MORBs (Schilling et al., 1983). Solid triangles = **N-MORB** ($(La/Sm)_N < 1.0$); Open squares = **E-MORB** ($(La/Sm)_N > 1.8$); Gray circles = **T-MORB** (Transitional MORB, $1.0 < La/Sm < 1.8$) forming a continuous hybrid mixing spectrum.

3. INCOMPATIBLE TRACE ELEMENT MULTI-ELEMENT DIAGRAMS & THE ORIGIN OF THE DEPLETED MANTLE

When trace elements are arranged on an **extended rare earth diagram (spidergram)** in order of increasing compatibility during mantle melting (from Cs, Rb, Ba, Th, U on the far left to Y, Yb, Lu on the far right), the incompatible element depletion of MORB is remarkably clear.

> *"Figure 11.18 compares the incompatible element abundances in MORB and typical OIB alkali basalt in a spider diagram, in which elements are ordered according to their incompatibility and normalized to primitive mantle values... **The incompatible element depletion of MORB is apparent.** OIB are generally incompatible-element enriched... In the view of most mantle geochemists and geophysicists, MORB are derived from the asthenosphere, which is that part of the upper mantle that directly underlies the lithosphere and is characterized by relatively low seismic velocities... Oceanic island basalts are thought to be derived from mantle plumes that rise from a thermal boundary layer at either the 660 km discontinuity or the core-mantle boundary."*
>
> **— William M. White, *Geochemistry*, Chapter 11 (The Mantle and Core), p. 491**

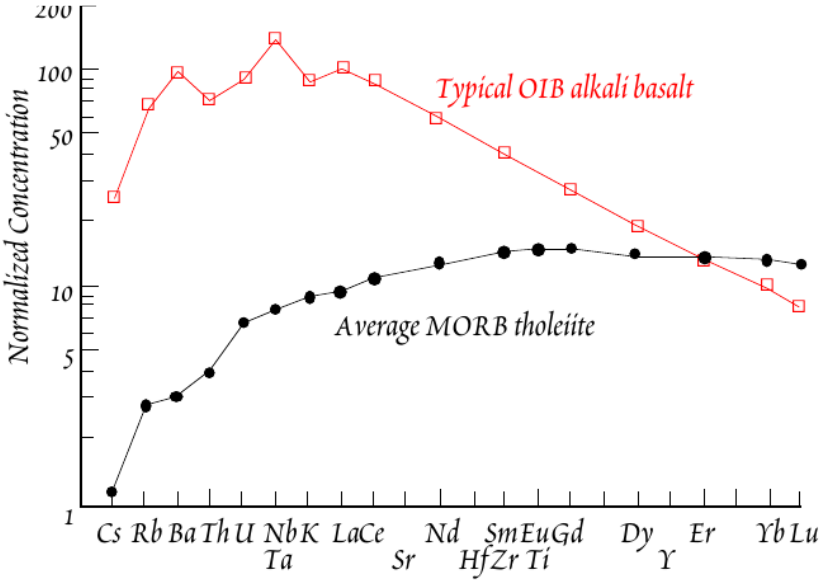


Figure 11.18. MORB tholeiite and OIB alkali basalt are compared in an extended rare earth plot or 'spider diagram' in which the elements are ordered with compatibility increasing to the right. Data are normalized to "primitive mantle" values. After Sun (1980).

MORB source through a OIB-source upper mantle

White Figure 11.18: Primitive mantle-normalized incompatible trace element spider diagram for Average MORB tholeiite compared with typical OIB alkali basalt (Sun, 1980). Note the profound depletion of MORB in the most incompatible elements (Cs, Rb, Ba, Th, U, Nb, Ta, K) dropping to near or below 1× primitive mantle, in stark contrast to OIB which reaches 50–150× primitive mantle.

The Hofmann (1988) Complementary Crust–Mantle Paradigm

Why is the upper mantle depleted in incompatible elements? The definitive geochemical answer was articulated by Albrecht Hofmann (1988): **the Depleted MORB Mantle (DMM) is the complementary residue left behind by the extraction of the Earth's continental crust**.

> *"Hofmann (1988) argued that the compositional relationship between continental crust and oceanic crust could be explained by a simple two-stage model of first extracting continental crust from the primitive mantle, and then extracting oceanic crust by remelting of the residual (depleted) mantle... **Fig. 12.15 shows complementary patterns exhibited by concentrations of elements in the bulk continental crust and 'normal-type' mid-ocean ridge basalts (N-MORB)**, both normalized to the estimated values for the primitive mantle (Sun and McDonough, 1989). The continental crust is enriched in incompatible elements... whereas N-MORB is depleted in these elements."*
>
> **— Kula C. Misra, *Introduction to Geochemistry: Principles and Applications*, Chapter 12, p. 308**

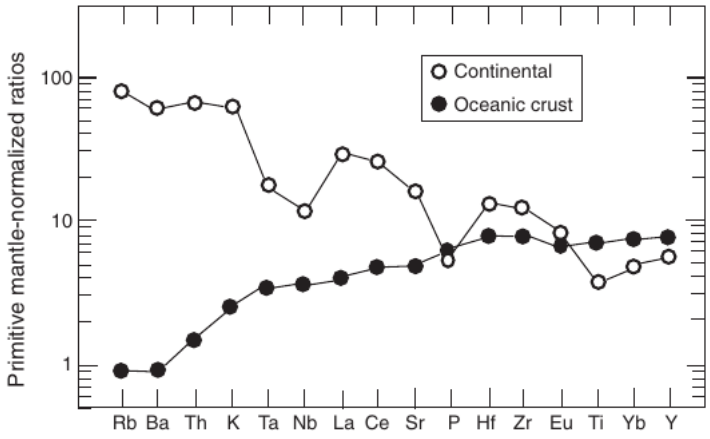


Fig. 12.15 Complementary patterns exhibited by concentrations of elements in the bulk continental crust (Condie, 1997) and "normal-type" mid-ocean ridge basalts (N-MORB) (Hofmann, 1988), both normalized to the estimated values for the primitive mantle (Sun and McDonough, 1989. The continental rust is enriched in incompatible elements and has a high La/Nb ratio, which are characteristic features of convergent margin magmas and rarely observed in intraplate magmas.

Misra Figure 12.15: Complementary trace element patterns of bulk continental crust (open circles) and N-MORB (solid circles), normalized to Primitive Mantle (Hofmann, 1988; Sun & McDonough, 1989). Incompatible elements strongly partitioned into the continental crust (yielding enrichment factors of 30–80×) leave the residual upper mantle severely depleted, directly explaining why N-MORB has normalized ratios < 1 for Rb, Ba, and Th.

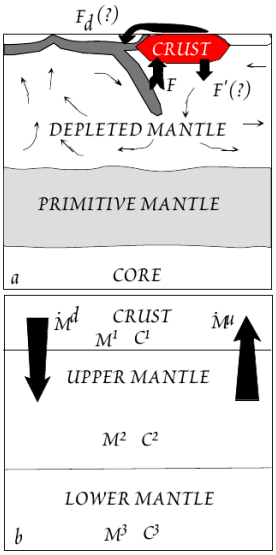


Figure 11.21. The three reservoir model of the mantle. The depleted mantle is the source of MORB and has ε_{Nd} = +10, the lower mantle is primitive and has bulk Earth characteristics,

> **White Figure 11.21:** The Three-Reservoir Model of the Earth's mantle and crust (DePaolo, 1980; White, 2005). The extraction of the continental crust (enriched in incompatible elements) leaves the **Depleted Mantle** (source of MORB, characterized by ε_{Nd} = +10 and low $^{87}Sr/^{86}Sr$), underlain by an undepleted Primitive Mantle or deep plume source.

4. RADIOGENIC ISOTOPE FINGERPRINTING OF MANTLE RESERVOIRS (DMM, BSE, HIMU, EMI, EMII)

Because radiogenic isotopic ratios (e.g., $^{87}Sr/^{86}Sr$, $^{143}Nd/^{144}Nd$) do not change during partial melting or fractional crystallization, they reflect the time-integrated parent/daughter ratio of the mantle source over hundreds of millions of years:

- **Rb–Sr System:** Rb is more incompatible than Sr ($D_{Rb} < D_{Sr}$). Extraction of continental crust depleted the upper mantle in Rb, leaving a low Rb/Sr ratio. Over 2–3 billion years, little radiogenic ^{87}Sr grew, so **DMM has very low $^{87}Sr/^{86}Sr$ (< 0.7025)**.
- **Sm–Nd System:** Nd is more incompatible than Sm ($D_{Nd} < D_{Sm}$). Continental crust extraction extracted more Nd than Sm, leaving the residual upper mantle with an *elevated Sm/Nd ratio*. Decay of ^{147}Sm to ^{143}Nd produced high radiogenic ^{143}Nd, so **DMM has high $^{143}Nd/^{144}Nd$ (> 0.5131) and positive ε_{Nd} (+10 to +12)**.

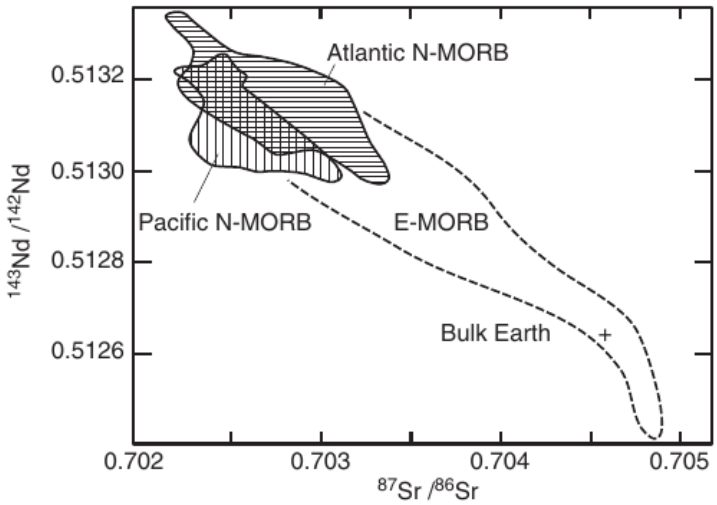


FIGURE 13 Sr and Nd isotopic ratios for Atlantic N-MORB (horizontal hatches), Pacific N-MORB (vertical hatches), and E-MORB of the southwestern Indian Ocean. Data from Ito et al. (1987), LeRoex et al. (1983), and Meyzen et al. (2005).

mantle source. Note that E-MORBs extend the MORB array to much more enriched values (higher $^{87}Sr/^{86}Sr$ and lower

Winter Figure 13-13: $^{143}Nd/^{144}Nd$ versus $^{87}Sr/^{86}Sr$ for Atlantic and Pacific N-MORBs compared with E-MORBs (Winter, 2010; Ito et al., 1987). N-MORB forms a tight cluster in the upper-left quadrant (high $^{143}Nd/^{144}Nd > 0.5130$, low $^{87}Sr/^{86}Sr < 0.7032$), characteristic of long-term incompatible element depletion. E-MORBs extend downward and rightward toward Bulk Earth (BSE) and enriched mantle components.

TABLE 5 Approximate Isotopic Ratios of Various Mantle Reservoirs

Reservoir	$^{87}Sr/^{86}Sr$	$^{144}Nd/^{143}Nd$	$^{206}Pb/^{204}Pb$	$^{207}Pb/^{204}Pb$	$^{3}He/^{4}He$	$^{187}Os/^{186}Os$	$^{18}O/^{16}O$
End-Member Mantle							
DM	0.7015–0.7025	0.5133–0.5136	15.5–17.7	<15.45	7–9 R_A	0.123–0.126	low
HIMU	0.7025–0.7035	0.511–0.5121	21.2–21.7	15.8–15.9	2–6 R_A?	0.15	low
EMI	c 0.705	<0.5112	17.6–17.7	15.46–15.49	2–6 R_A?	0.152	low
EMII	>0.722	0.511–0.512	16.3–17.3	15.4–15.5	2–6 R_A?	0.156?	high
FOZO*	0.7030–0.7040	0.51280–0.51300	18.50–19.50	15.55–15.65	8–32 R_A	low	low
Other Mantle							
BSE	0.7052	0.51264	18.4	15.58	40–80 R_A?	0.129	low
PREMA	0.7033	<0.5128	18.2–18.5	15.4–15.5			
PHEM	0.7042–0.7052	0.51265–0.51280	18.4–19.0	15.5–15.6	>35 R_A	low	low
C	0.703–0.704	0.5128–0.5129	19.2–19.8	15.55–15.65	20–25+?	low	low
Continental Crust**	0.72–0.74	0.507–0.513	up to 28	up to 20	~0.1 R_A	high	high

*FOZO or some low-^{4}He reservoir, such as PHEM or C.

**Continental crust may be recycled back into the mantle via subduction.

Winter Table 14-5: Approximate isotopic ratios of principal mantle reservoirs (Rollinson, 1993; Winter, 2010). Note that **DM / DMM (Depleted Mantle)** exhibits the lowest $^{87}Sr/^{86}Sr$ (0.7015–0.7025) and highest $^{143}Nd/^{144}Nd$ (0.5133–0.5136), defining the source of **N-MORB**.

5. DECOMPRESSION MELTING MECHANICS & PLUME–RIDGE INTERACTION

Melting beneath mid-ocean ridges is driven by **adiabatic decompression** of upwelling mantle peridotite (lherzolite/harzburgite). Because the dry peridotite solidus has a steeper dP/dT slope than the mantle adiabat (~0.3°C/km), upward flow causes the mantle to cross its solidus at depths of ~60–80 km (P ≈ 1.5–2.5 GPa):

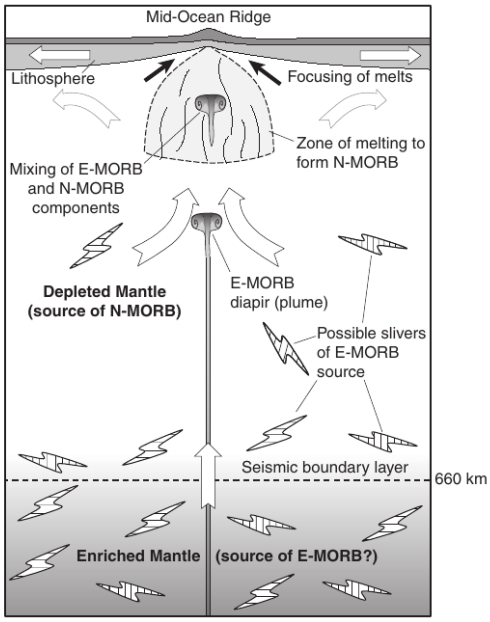


FIGURE 14 Schematic diagram illustrating a model for the rise and decompression melting of different source regions to produce N-type and E-type MORBs. After Zindler et al. (1984) and Wilson (1989). The gradational nature of the 660-km discontinuity and multiple slivers of enriched source material are intended to represent the uncertain nature of mantle

> **Winter Figure 13-14:** Schematic model for the rise and decompression melting of different mantle source regions beneath mid-ocean ridges (after Zindler et al., 1984; Wilson, 1989; Winter, 2010). The shallow, convecting upper mantle above 660 km consists of **Depleted Mantle (source of N-MORB)**. Deep mantle plumes ascending from the lower mantle tap **Enriched Mantle (source of E-MORB/P-MORB)**. Where plumes interact with spreading ridges (e.g., Iceland on MAR), melt mixing produces **T-MORB**.

Melting proceeds by *incremental polybaric near-fractional melting*, where small melt fractions (~1%) segregate continuously as the mantle rises through a triangular melting regime (Langmuir et al., 1992), focusing into the axial magma chamber to form the oceanic crust.

6. DISTINCT PETROGENETIC SETTING OF VOLCANIC ARC BASALTS (VAB)

Option (d) in the question refers to **VAB (Volcanic Arc Basalts)**. It is essential to recognize that VABs are completely distinct from Mid-Ocean Ridge Basalts in tectonic regime, melting mechanism, and geochemical signature:

- **Tectonic Setting:** VAB erupts at convergent plate boundaries (island arcs and active continental margins above subduction zones), whereas MORB erupts at divergent spreading ridges.
- **Melting Trigger:** MORB is generated by *dry adiabatic decompression melting*. VAB is generated by *hydrous flux melting*: aqueous fluids released by the dehydration of subducting serpentinite, altered basalts, and pelagic sediments lower the solidus of the overlying mantle wedge.
- **Geochemical Signature:** Subduction fluids selectively transport soluble **Large Ion Lithophile Elements (LILE: Cs, Rb, Ba, Sr, Pb)** into the mantle wedge, while **High Field Strength Elements (HFSE: Nb, Ta, Zr, Hf, Ti)** are retained in residual refractory titanate phases (such as rutile) in the subducting slab. This produces the universally diagnostic **negative Nb–Ta anomaly** and **positive Pb anomaly** in VAB, which is entirely absent in MORB.

PART II | DGMS 2019 Examination: Question & Worked Solution

Exhaustive option-by-option critical appraisal and definitive textbook-grounded answer

ORIGINAL QUESTION — DGMS 2019 (JUNIOR SCIENTIFIC OFFICER)

Q.25. Which one of the following Mid-Oceanic Ridge Basalts (MORB) is derived from an incompatible element-poor mantle?

- **(a) N-MORB**
- (b) P-MORB
- (c) T-MORB
- (d) VAB

Source: DGMS Junior Scientific Officer Exam 2019, Question 25. Reference Screenshot: `Screenshot from 2026-10-12 23-14-41.png`.

✓ CORRECT OPTION: (a) N-MORB

N-MORB ("Normal" MORB) is derived from the Depleted MORB Mantle (DMM), an incompatible element-poor upper mantle reservoir left behind after ancient extraction of continental crust.

EXHAUSTIVE OPTION-BY-OPTION CRITICAL ANALYSIS

(a) N-MORB (Normal Mid-Ocean Ridge Basalt) — CORRECT

Why it is correct: "N" stands for *Normal*. N-MORB constitutes the vast majority (~85%) of volcanic rock generated along oceanic spreading ridges globally. It is derived by adiabatic decompression partial melting of the **Depleted MORB Mantle (DMM)**—the convecting upper asthenospheric mantle above ~660 km depth.

- **Trace Element Depletion:** The DMM experienced previous melt extraction events (most notably the progressive formation of the continental crust throughout the Archean and Proterozoic). Because incompatible elements preferentially enter the liquid phase during melting, the residual DMM was left severely depleted in highly incompatible trace elements (Cs, Rb, Ba, Th, U, Nb, Ta, K, La, Ce).
- **Rare Earth Fingerprint:** N-MORB possesses a characteristic *positive LREE slope* on chondrite-normalized diagrams, with normalized **$(La/Sm)_N < 1.0$ (typically < 0.7)**, demonstrating that the source had already lost its light rare earths.
- **Isotopic Depletion:** High time-integrated Sm/Nd and low Rb/Sr in the depleted mantle result in highly depleted isotopic compositions: low $^{87}Sr/^{86}Sr$ (< 0.7025) and high $^{143}Nd/^{144}Nd$ (> 0.5131, ε_{Nd} = +10 to +12).

Winter (2010, p. 300) explicitly writes: *"We can thus distinguish between N-MORB ('normal' MORB), which taps the depleted, or incompatible-poor mantle, and E-MORB ('enriched' MORB, also called P-MORB for 'plume'), which taps an incompatible-richer mantle."* Thus, (a) is unequivocally the correct answer.

(b) P-MORB (Plume-type Mid-Ocean Ridge Basalt / E-MORB) — INCORRECT

Why it is incorrect: "P" stands for *Plume* (also termed **E-MORB** for *Enriched* MORB). P-MORBs occur along ridge segments that are situated near or directly influenced by deep-seated mantle plumes and hotspots (e.g., the Mid-Atlantic Ridge near Iceland, the Azores, or Tristan da Cunha, and the Galapagos Spreading Center).

P-MORBs are derived from an **incompatible element-rich (fertile or metasomatized) mantle source**, not an incompatible element-poor mantle. They have elevated concentrations of K_2O (>0.10 wt.%), TiO_2 (>1.0 wt.%), P_2O_5, and enriched LREE profiles with negative slopes where **$(La/Sm)_N > 1.8$**. Their radiogenic isotopes plot toward enriched reservoirs (higher $^{87}Sr/^{86}Sr$ ~ 0.7030–0.7045 and lower $^{143}Nd/^{144}Nd$). Therefore, P-MORB represents the opposite of an incompatible element-poor melt.

(c) T-MORB (Transitional Mid-Ocean Ridge Basalt) INCORRECT

Why it is incorrect: "T" stands for *Transitional*. T-MORBs exhibit geochemical characteristics intermediate between N-MORB and E-MORB/P-MORB, with **$(La/Sm)_N$ ratios typically ranging between 0.8 and 1.8**.

As documented by Schilling et al. (1983) and Winter (2010, p. 301), chemical and isotopic data demonstrate that T-MORBs do not represent a unique, distinct third mantle layer, but are formed by the **mixing of depleted N-MORB melts with enriched E-MORB plume melts** during magma ascent and storage in axial magma chambers. Because T-MORBs involve significant contributions from the enriched mantle component, they are not derived from a purely incompatible element-poor mantle source.

(d) VAB (Volcanic Arc Basalts) INCORRECT

Why it is incorrect: VAB stands for *Volcanic Arc Basalts*. This option is a double category error:

1. **Not a MORB:** VAB is erupted at **convergent plate boundaries (subduction zones)**, such as island arcs and continental volcanic arcs (e.g., the Marianas, Aleutians, Andes). The question specifically asks for a *Mid-Oceanic Ridge Basalt (MORB)*.
2. **Not incompatible element-poor:** VABs are strongly enriched in Large Ion Lithophile Elements (LILE: Cs, Rb, Ba, Sr, Pb) and LREE derived from hydrous slab fluids fluxing the mantle wedge. Furthermore, they display a profound depletion in HFSE (Nb, Ta, Ti), producing the classic subduction-zone negative Nb–Ta anomaly.

PART III

Synthesis Reference Tables & High-Yield Practice Questions

Comprehensive diagnostic reference tables and 8 curated exam questions for DGMS, UPSC GSI, GATE & IIT-JAM

MASTER REFERENCE TABLES: OCEANIC BASALT SYSTEMATICS & MANTLE RESERVOIRS

Table 1: Comprehensive Comparison of MORB Varieties and Oceanic Island Basalts (OIB)

Geochemical Class	Mantle Source Reservoir	$(La/Sm)_N$	REE Slope	K_2O (wt. % at Mg# 65)	$^{87}Sr/^{86}Sr$	$^{143}Nd/^{144}Nd$	Tectonic Setting
N-MORB (Normal)	DMM (Depleted MORB Mantle)	< 1.0 (0.4–0.8)	Positive (LREE depleted)	< 0.10	0.7020 – 0.7028	> 0.5131 (ε_{Nd} +10 to +12)	Spreading ridges away from plumes
T-MORB (Transitional)	Hybrid / Mixed (DMM + Enriched Plume)	0.8 – 1.8	Flat to slightly sloped	0.10 – 0.25	0.7027 – 0.7032	0.5129 – 0.5131 (ε_{Nd} +6 to +9)	Transition zones along ridge axes
E-MORB / P-MORB (Enriched / Plume)	Enriched Mantle / Plume Diapir	> 1.8 (up to 5.0)	Negative (LREE enriched)	> 0.15 (up to 1.5)	0.7030 – 0.7045	0.5127 – 0.5129 (ε_{Nd} +2 to +6)	Ridge-plume intersections (Iceland, Azores)
OIB (Ocean Island Basalt)	Deep Plumes (HIMU, EMI, EMII, FOZO)	> 2.0 (up to 10.0)	Steep negative	0.5 – 3.5	0.7030 – 0.7070+	0.5124 – 0.5130 (ε_{Nd} -2 to +8)	Intraplate oceanic hotspots (Hawaii, Samoa)
VAB (Volcanic Arc Basalt)	Subduction-fluid fluxed mantle wedge	Variable (high LILE/HFSE)	Negative Nb-Ta anomaly	0.3 – 2.0	0.7035 – 0.7060+	0.5127 – 0.5130	Convergent subduction boundaries

Table 2: Key Sub-Oceanic Mantle Reservoirs Defined by Radiogenic Isotopes (Zindler & Hart, 1986)

Reservoir Code	Full Name	$^{87}Sr/^{86}Sr$	$^{143}Nd/^{144}Nd$	Proposed Petrogenetic Origin & Significance
DMM (DM)	Depleted MORB Mantle	0.7015 – 0.7025	0.5133 – 0.5136	Residue left after extraction of the continental crust; source of **N-MORB**.
BSE (PM)	Bulk Silicate Earth / Primitive Mantle	~0.7045 – 0.7052	0.51264 ($\varepsilon_{Nd} = 0$)	Theoretical pristine composition of the silicate Earth before crust-mantle differentiation.
HIMU	High-μ (High $^{238}U/^{204}Pb$)	0.7028 – 0.7035	0.5128 – 0.5130	Subducted ancient altered oceanic crust where hydrothermal alteration removed Pb relative to U.
EMI	Enriched Mantle Type I	~0.7050	< 0.5124	Recycled pelagic sediments or subcontinental lithospheric mantle (SCLM) delamination.
EMII	Enriched Mantle Type II	> 0.7070	0.5120 – 0.5124	Recycled upper continental crust or terrigenous sediments carrying high Rb/Sr.
FOZO / PREMA	Focal Zone / Prevalent Mantle	0.7030 – 0.7040	0.5128 – 0.5130	Ubiquitous common mixing component in deep mantle plumes.

Table 3: Diagnostic Trace Element Ratios for Tectonic Discrimination of Basalts

Ratio	N-MORB	E-MORB	OIB	Volcanic Arc Basalts (VAB)	Continental Crust
$(La/Sm)_N$	< 1.0 (0.4–0.8)	> 1.8	> 2.5	Variable (1.2–3.0)	3.0 – 5.0
Nb / U	47 ± 10	47 ± 10	47 ± 10	< 10 (pronounced deficit)	~10
Ce / Pb	25 ± 5	25 ± 5	25 ± 5	< 5 (Pb enrichment)	~4
Zr / Nb	> 30 (typically 30–50)	10 – 20	< 10 (typically 4–8)	Variable	~10
Ba / Nb	< 5	~8	~10	> 30 (up to 100+)	~50

HIGH-YIELD CURATED PRACTICE QUESTIONS (DGMS, UPSC GSI, GATE & IIT-JAM)

Practice Question 1 (UPSC Geologist 2022 Pattern):

Which of the following rare earth element ratios is globally utilized as the primary diagnostic slope index to discriminate between LREE-depleted N-MORB and LREE-enriched E-MORB?

Answer: $(La/Sm)_N$ (Chondrite-normalized Lanthanum-to-Samarium ratio).

Explanation: Because HREEs (Tb to Lu) have virtually overlapping, flat concentrations in both N-MORB and E-MORB (controlled by garnet/spinel melting physics), the overall slope ratio $(La/Lu)_N$ or $(La/Yb)_N$ can be misleading if residual garnet is involved. In contrast, $(La/Sm)_N$ directly measures the slope across the light rare earth elements where incompatible element fractionation is most pronounced: $(La/Sm)_N < 1.0$ defines N-MORB, while $(La/Sm)_N > 1.8$ defines E-MORB.

Practice Question 2 (GATE Geology 2021 Pattern):

In a diagram of $^{143}Nd/^{144}Nd$ versus $^{87}Sr/^{86}Sr$, the Depleted MORB Mantle (DMM) source plots in which quadrant relative to the Bulk Silicate Earth (BSE)?

Answer: Upper-left quadrant (high $^{143}Nd/^{144}Nd$, low $^{87}Sr/^{86}Sr$).

Explanation: Continental crust formation extracted incompatible elements over 3.8+ Ga, depleting the upper mantle in Rb relative to Sr (low Rb/Sr) and in Nd relative to Sm (high Sm/Nd). Over geologic time, low Rb/Sr yielded minimal ^{87}Sr accumulation ($^{87}Sr/^{86}Sr < 0.7025$), whereas high Sm/Nd yielded substantial ^{143}Nd accumulation ($^{143}Nd/^{144}Nd > 0.5131$, $\varepsilon_{Nd} = +10$ to $+12$).

Practice Question 3 (IIT-JAM Geology Pattern):

According to the complementary crust–mantle evolution model of Hofmann (1988), which planetary process was primarily responsible for depleting the N-MORB mantle source in incompatible elements?

Answer: The irreversible, episodic extraction of the continental crust from the primitive mantle.

Explanation: Incompatible trace elements (Cs, Rb, Ba, Th, U, K, LREE) have bulk solid/liquid distribution coefficients $D << 1$. During partial melting, they strongly partition into the magma. Because low-density, buoyant felsic continental crust cannot be subducted back into the convective mantle, its growth sequestered these elements, leaving the upper convecting asthenosphere as a depleted, incompatible-poor residue (DMM).

Practice Question 4 (UPSC GSI Mains Analytical Question):

Explain how Volcanic Arc Basalts (VAB) can be distinguished from Mid-Ocean Ridge Basalts (MORB) on multi-element spidergrams using High Field Strength Elements (HFSE).

Answer: Presence of a marked negative Nb–Ta anomaly and positive Pb anomaly in VAB.

Explanation: In subduction zones, aqueous fluids from the slab transport water-soluble LILE (Cs, Rb, Ba, Sr, Pb) into the mantle wedge, but cannot transport insoluble HFSE (Nb, Ta, Zr, Hf, Ti), which remain bound in residual titanate minerals (rutile) in the slab. Consequently, VAB displays very high Ba/Nb (>30) and low Nb/U (<10), manifesting as a sharp trough at Nb–Ta on spidergrams. In MORBs (both N-MORB and E-MORB), no negative Nb–Ta anomaly exists (Nb/U ≈ 47).

Practice Question 5 (DGMS Mining / Geology 2018 Pattern):
What is the primary physical mechanism of mantle melting that produces parental magmas at mid-ocean spreading centers?

Answer: Decompression melting (adiabatic upwelling of mantle peridotite).

Explanation: Plates pull apart at divergent boundaries, causing underlying asthenospheric mantle peridotite to ascend passively. Because the ascent is rapid relative to conductive thermal loss, the path is adiabatic (~0.3°C/km). The solidus slope (dP/dT) of peridotite is much steeper than the adiabat, so the upwelling rock intersects the dry solidus at depths between 60 and 80 km, generating basaltic melts without any external heat or water addition.

Practice Question 6 (Petrology Competitive Exam):
Along the Mid-Atlantic Ridge near Iceland, basalts with flat REE patterns and intermediate incompatible element abundances ($(La/Sm)_N \approx 1.0–1.6$) are classified as:

Answer: T-MORB (Transitional MORB).

Explanation: Where deep mantle plumes impinge on oceanic spreading centers (e.g., the Iceland hotspot on the Mid-Atlantic Ridge), ascending enriched plume diapir melts mix with ambient depleted N-MORB melts within sub-ridge melting regimes and shallow magma chambers, producing hybrid transitional lavas termed T-MORBs.

Practice Question 7 (Isotope Geochemistry Specialist Exam):
Why do both N-MORB and OIB share identical canonical ratios of Nb/U = 47 ± 10 and Ce/Pb = 25 ± 5, despite vastly different absolute abundances?

Answer: Because Nb and U (and Ce and Pb) have identical bulk partition coefficients during purely oceanic mantle melting and differentiation.

Explanation: Hofmann et al. (1986) demonstrated that $D_{Nb} \approx D_U$ and $D_{Ce} \approx D_{Pb}$ during igneous processes in the oceanic mantle. Fractionation only occurs when continental crust or subduction fluids are involved, where aqueous fluids fractionate Pb from Ce (producing Ce/Pb < 5 in arcs and crust) and rutile fractionates Nb from U (producing Nb/U < 10 in continental crust).

Practice Question 8 (Advanced Igneous Petrology):
During fractional crystallization of a parental MORB magma in a shallow crustal axial magma chamber, which pair of trace elements will decrease rapidly in the liquid?

Answer: Nickel (Ni) and Chromium (Cr).

Explanation: Ni has a very high partition coefficient in olivine ($D_{Ni}^{ol/liq} >> 1$, typically 5–15), and Cr is strongly compatible in magnesiochromite and clinopyroxene ($D_{Cr} >> 1$). Consequently, early fractional crystallization of olivine + spinel immediately depletes the derivative MORB melt in Ni and Cr, while incompatible elements (Zr, Ti, Y, K) steadily increase.